

3.3.4.

Even 4 digit number n , all divisors of n increasing order 323 is one of the numbers, m is to the immediate right. min value of m . We know that $323 \mid n$, and there exists another $m \mid n$.

$323 = 17 \cdot 19$, so the prime factorization of n must go something like $n = 17 \cdot 19 \dots$

$\frac{9999}{323}$ is approximately 31, so any additional factor(s) must be less than 31 combined

$m > 323$ but we are also minimizing m . The next largest factor we can include is 20, so $n = 17 \cdot 19 \cdot 20$.

We can combine this factorization to create the next largest factor, which must be $\boxed{17 \cdot 20 = 340}$

3.4.4.

$$(63 \bmod a) + (91 \bmod a) + (129 \bmod a) = 25$$

We can add over moduli, so $63 + 91 + 129 \equiv 25 \pmod{a}$

$$283 \equiv 25 \pmod{a}, \text{ so } 283 - 25 \equiv 0 \pmod{a}$$

$a \mid 258$ and $a > 8$ since $\frac{25}{3} > 8$ so cannot have all less than 8, $258 = 2 \cdot 3 \cdot 43$

$\boxed{a = 43}$ works by a quick check.